

LIGHTS IN THE SKY

34–37 Radiation

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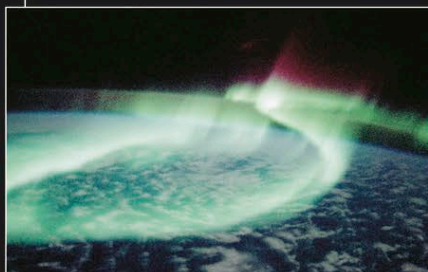
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AS WELL AS STARS, GALAXIES, NEBULAE, and solar system objects, other phenomena can cause lights to appear in the night sky. Mainly, these originate in light or particles of matter reaching Earth in various indirect ways from the Sun, but some are generated by Earth-bound processes. Amateur stargazers need to be aware of these sources of nocturnal light to avoid confusion with astronomical phenomena.



AURORA FROM THE SPACE SHUTTLE

This photograph of the aurora australis was taken from the Space Shuttle Discovery during a 1991 mission. A study of the aurora's features was one of the mission tasks.

AURORAE

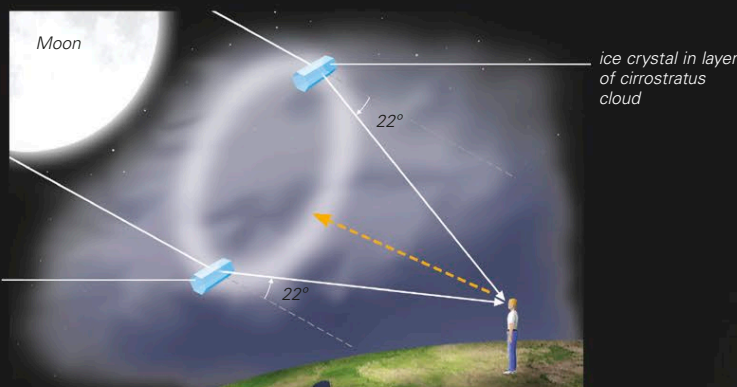
The aurora borealis (northern lights) and aurora australis (southern lights) appear when charged particles from the Sun, carried to Earth in the solar wind (see pp.106–107), become trapped by Earth's magnetic field. They are then accelerated into regions above the north and south magnetic poles, where they excite particles of gas in the upper atmosphere, 60–250 miles (100–400 km) above Earth's surface. The appearance and location of aurorae change in response to the solar wind. They are most often visible at high latitudes, toward Earth's magnetic poles, but may be seen at lower latitudes during disturbances in the solar wind, such as after mass ejections from the Sun (see pp.106–107).

AURORA BOREALIS

A colorful display of the northern lights is visible here over silhouetted trees near Fairbanks, Alaska. The colors stem from light emission by different atmospheric gases.

ICE HALOES

Atmospheric haloes are caused by ice crystals high in Earth's atmosphere refracting light. Light either from the Sun or the Moon (that is, reflected sunlight) can cause haloes. The most common halo is a circle of light with a radius of 22° around the Moon or Sun. Also present may be splashes of light, called moon dogs or sun dogs (parhelia), arcs, and circles of light that seem to pass through the Sun or Moon. All these phenomena result from the identical angles between the faces of atmospheric ice crystals. Even if the crystals are not all aligned, they tend to deflect light in some directions more strongly than in others.



OBSERVING A 22° HALO

This halo is formed when ice crystals in the atmosphere refract light from the Moon to the observer on Earth by an angle of 22° . A light ray is refracted through this angle as it passes through two faces of an ice crystal.



HALO AND MOON DOGS

This photograph taken in Arctic Canada shows several refraction phenomena. The patches of light on either side of the Moon, called moon dogs, are caused by horizontal ice crystals in the atmosphere refracting light. The band of light running through the moon dogs is called a paraselenic circle. Also visible is a circular 22° halo.